Anomaly Detection in Industrial Equipment Sensor Data

Assignment ID: ASGN2703 | **Method:** Isolation Forest | **Data:** industrial_sensor_data.csv (3,000 records, 10 columns)

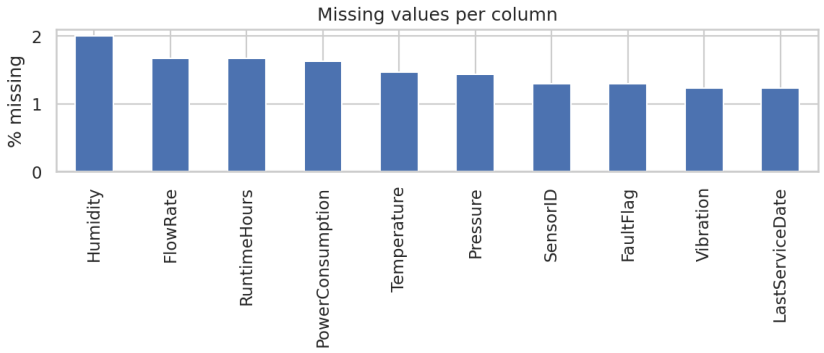
1. Executive summary

An Isolation Forest was trained on eight standardised features (seven sensor channels plus a derived service-age feature) to detect anomalous equipment readings. The model works as designed: it ranks readings by how unusual they are and flags 147 of 2961 labelled records at 5% contamination. **However, the flagged readings do not line up with the dataset's** `FaultFlag` **label** (precision 0.41, recall 0.04, F1 0.08, ROC-AUC 0.48). A supervised Random Forest also fails to predict the label (cross-validated ROC-AUC 0.50), and no sensor correlates with it (max |r| = 0.03). We therefore conclude that, in this dataset, `FaultFlag` carries no signal recoverable from the sensor readings, so the weak scores reflect the data rather than a defect in the model. The key recommendation is to validate how `FaultFlag` is defined and recorded, and to use the anomaly score as a prioritised inspection list whose outcomes become trustworthy labels.

2. Dataset and preprocessing

The dataset contains 3,000 records: SensorID, Temperature, Pressure, Vibration, Humidity, FlowRate, PowerConsumption, RuntimeHours, LastServiceDate and the binary label FaultFlag (47.6% faulty among labelled rows). Preprocessing steps:

- **Missing values.** 448 cells (422 rows) were missing, about 1-2% per column. Sequential SensorIDs were reconstructed from row position (verified on all known IDs); missing service dates were filled from neighbouring rows (the data is date-ordered); sensor values were median-imputed (robust to outliers); the label was never imputed, and unlabelled rows were excluded from the metrics only.
- **Feature engineering.** LastServiceDate was converted to DaysSinceService relative to the latest date in the data.
- **Noise / extremes.** The IQR rule found no outliers in any feature; mild winsorising at the 0.5/99.5 percentiles (210 values) limited the influence of isolated spikes. A rolling median (window 25) was used for plotting only.
- **Normalisation.** Z-score standardisation of all features.



3. Model and evaluation

Model. scikit-learn IsolationForest with 300 trees, max_samples='auto', contamination = 0.05 (a conventional anomaly rate, as the true rate is unknown) and a fixed random seed (42). Anomalies receive shorter average path lengths and hence higher anomaly scores.

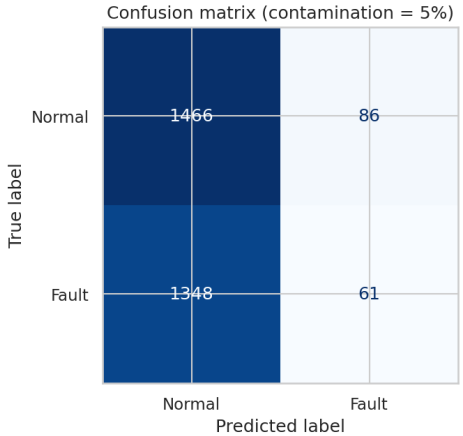
Metrics. FaultFlag = 1 was treated as the ground-truth 'anomaly'. Precision, recall and F1 depend on the threshold; ROC-AUC uses the continuous score.

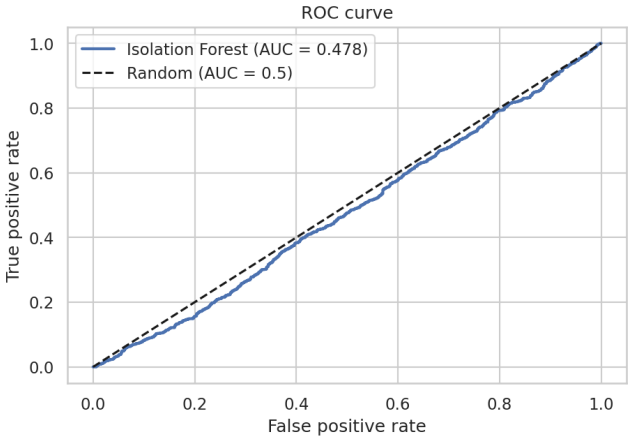
Metric	Value
Precision	0.415
Recall	0.043
F1-score	0.078
ROC-AUC	0.478
Confusion matrix [[TN, FP],[FN, TP]]	[[1466, 86], [1348, 61]]

Sensitivity to the contamination parameter

Contamination	Flagged	Precision	Recall	F1
0.010	29	0.379	0.008	0.015
0.050	147	0.415	0.043	0.078
0.100	297	0.424	0.089	0.148
0.200	590	0.425	0.178	0.251
0.300	882	0.446	0.279	0.343
0.476	1407	0.463	0.462	0.462

Precision stays at 0.38-0.46, essentially the 48% base rate, at every threshold, and F1 rises only because more points are flagged. Even when contamination equals the fault rate the model reaches F1 = 0.46, which is what random guessing achieves. The supervised sanity check (Random Forest, 5-fold CV ROC-AUC = 0.497 ± 0.011) confirms that the features contain no usable information about the label.





4. Visualisation of results

Rows are ordered by SensorID (and service date), so record order serves as a time axis. Red points are readings flagged as anomalous.

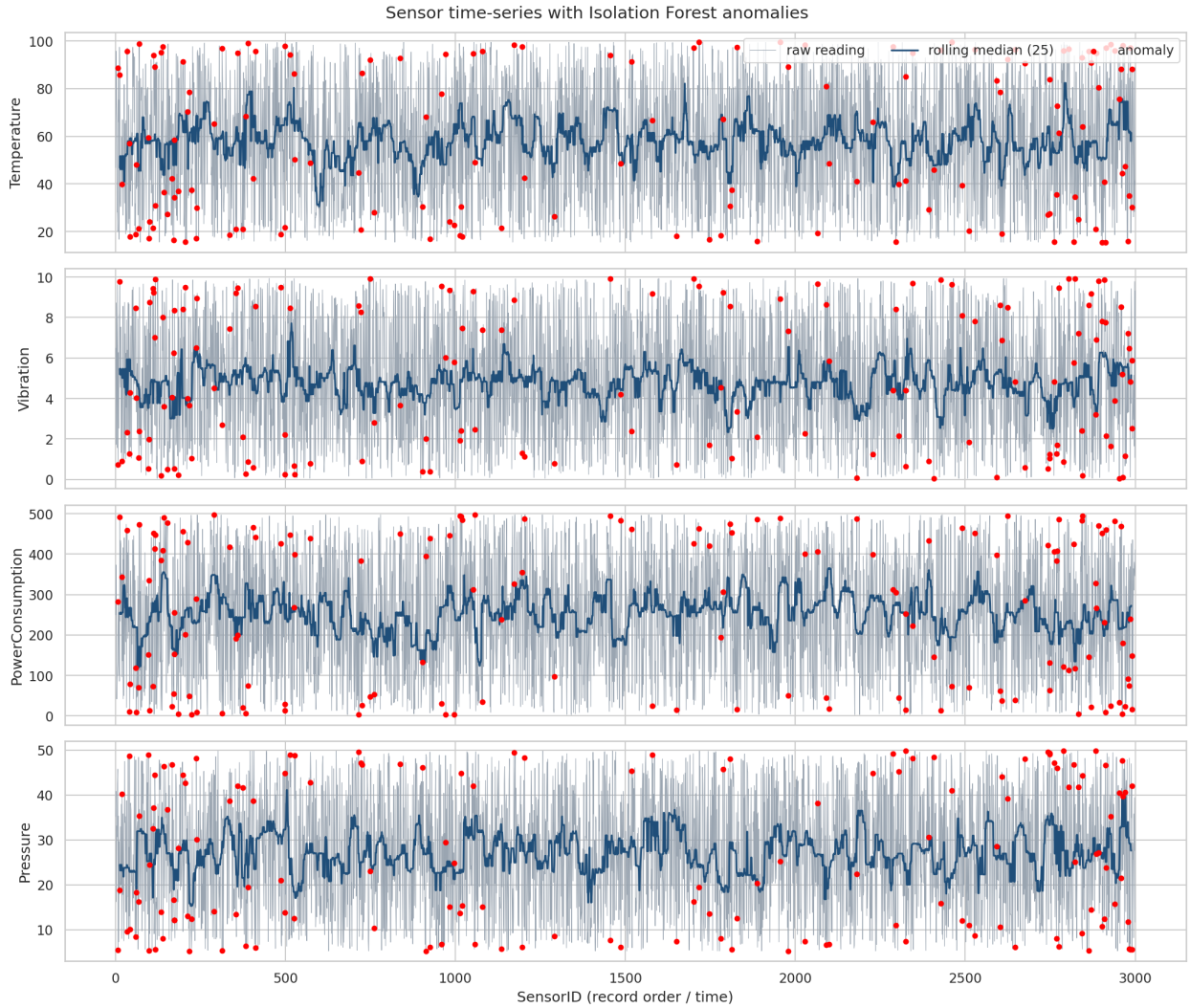


Figure: sensor time-series with rolling median and flagged anomalies.

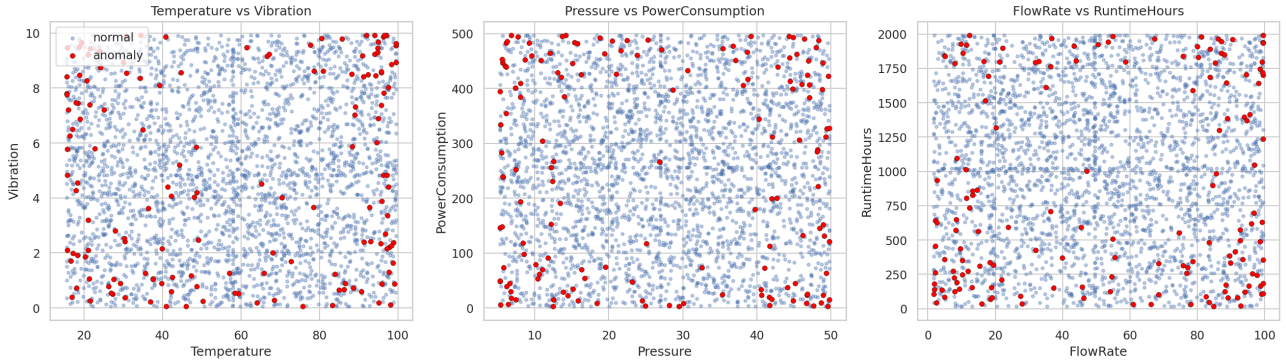


Figure: normal (blue) vs anomalous (red) readings for three sensor pairs.

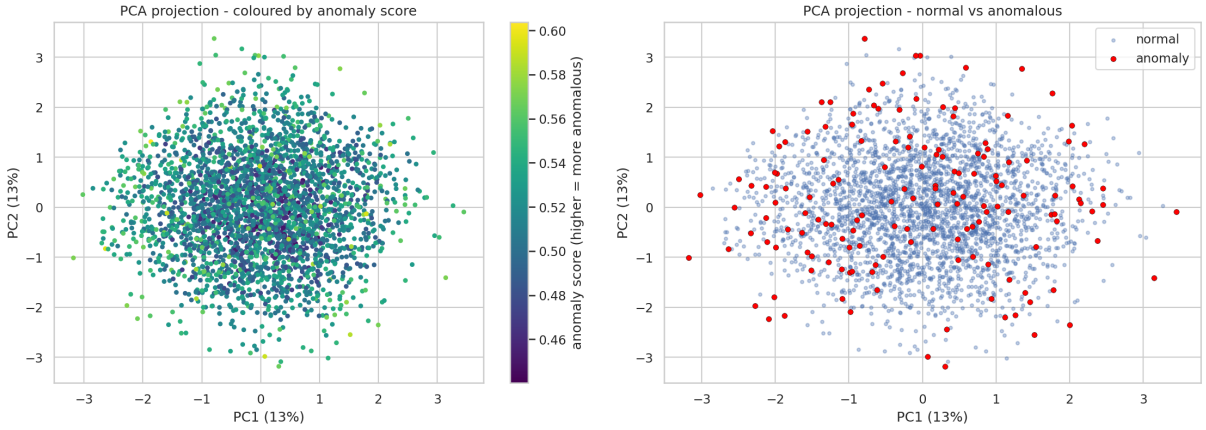
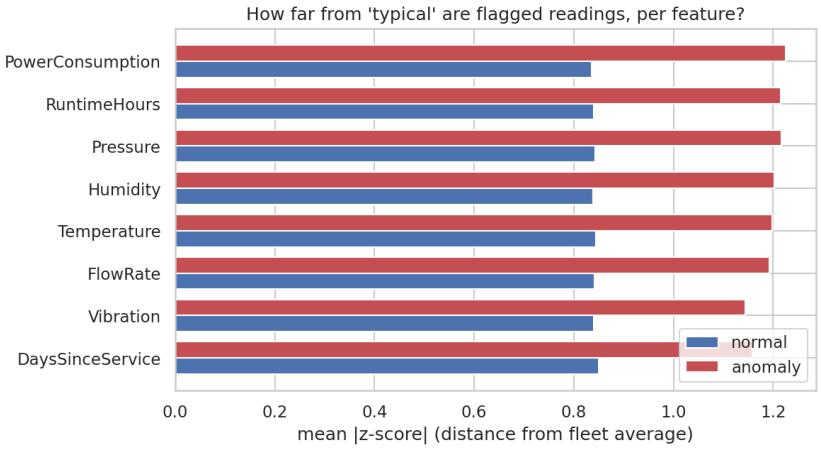


Figure: 2-D PCA projection coloured by anomaly score and by class.

Interpretation

The sensor features are close to uniformly distributed, independent of one another, and contain no IQR outliers. Isolation Forest therefore flags readings in the *corners* of the operating envelope: combinations in which several features are simultaneously far from typical. Flagged readings have on average 2.8 features with |z| > 1.5 compared with 1.0 for normal readings, and are farther from the fleet average on every feature, with no single sensor dominating. The only temporal pattern is at the edges: about 13% of records in the first and last 250 are flagged versus 1-5% in between. DaysSinceService falls steadily from 124 to 0 days across the record order, so the longest-unserviced and most recently serviced units sit at the extremes of that feature and are isolated more easily (25% of flagged units have a service age of 5 days or less, or 120 days or more, versus 8% of normal units).



5. Potential causes of anomalies

The data alone cannot establish causation, so the following are engineering hypotheses to verify during inspection:

Signal pattern	Plausible cause
High vibration + high temperature	Bearing wear, misalignment, imbalance, lubrication failure
High pressure with low flow rate	Blockage, fouled filter, partly closed valve
Low pressure with high flow rate	Leak, seal or pipe failure, pump cavitation
High power consumption at normal output	Motor inefficiency, friction, electrical fault
High temperature + high humidity	Cooling / ventilation failure, condensation
Long time since service + abnormal readings	Overdue maintenance, accumulated wear
Isolated single-sensor spikes	Sensor drift, calibration error, loose wiring (measurement fault, not equipment fault)

6. Recommendations for maintenance and process optimisation

- **Validate FaultFlag first.** A ~48% fault rate that no sensor explains suggests the flag is defined differently from what the sensors measure (e.g. an inspection or work-order flag) or is mis-recorded. Confirm its definition with the maintenance team before building models on it.
- **Use the anomaly score for triage.** Inspect the top-scoring units (listed in anomaly_results.csv) and record the findings. Confirmed outcomes become reliable labels for supervised models.
- **Check instrumentation.** Cross-check flagged readings against neighbouring sensors and recalibrate channels showing isolated spikes.
- **Adopt condition-based maintenance.** Set joint alert thresholds on vibration, temperature and power consumption, and prioritise service for units that are both long overdue and flagged.
- **Improve data quality.** Monitor the collection pipeline for the 1-2% missing values per column and record true timestamps so drift and trends can be modelled.
- **Operationalise.** Re-fit the model periodically, set contamination (e.g. 2-5%) according to inspection capacity, and track the precision of reviewed alerts over time.

7. Limitations

Contamination is a tuning choice; record order is only a proxy for time; and the dataset appears to contain no real fault structure (uniform, independent features), so the causes in Section 5 are hypotheses rather than findings.

Deliverables: anomaly_detection_isolation_forest.ipynb (code, comments, visualisations), this report, README.md, requirements.txt, anomaly_results.csv.